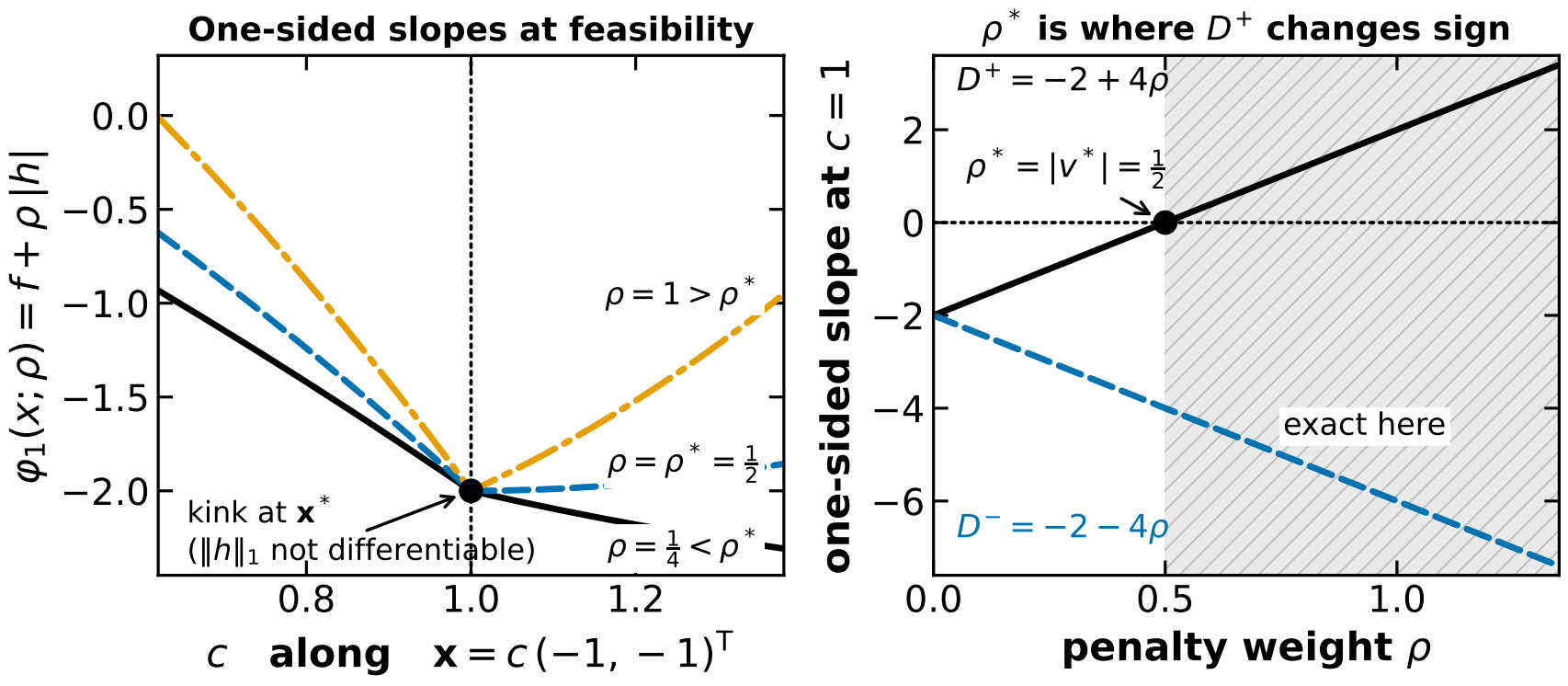
One-sided slopes at feasibility
$\varphi_1(x;\rho) = f + \rho\,|h|$
0.0
−0.5
−1.0
−1.5
−2.0
$\rho = 1 > \rho^*$
$\rho = \rho^* = \frac{1}{2}$
$\rho = \frac{1}{4} < \rho^*$
kink at $\mathbf{x}^*$
($\|h\|_1$ not differentiable)
0.8
1.0
1.2
c along $\mathbf{x} = c\,(-1, -1)^{\mathsf{T}}$
ρ^* is where D^+ changes sign
$D^+ = -2 + 4\rho$
$\rho^* = |v^*| = \frac{1}{2}$
exact here
$D^- = -2 - 4\rho$
one-sided slope at $c = 1$
2
0
−2
−4
−6
0.0
0.5
1.0
penalty weight ρ